

TANISHK JAIN

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Summary

B.Tech Computer Science (AI & ML) student with experience across machine learning, computer vision, and full stack development. Built end-to-end deep learning pipelines for medical image segmentation and deployed production web applications.

Experience

Indian Institute of Technology Delhi

June 2026 – September 2026

Machine Learning Research Intern

Delhi, India

- Developed end-to-end deep learning pipelines for multimodal retinal vessel segmentation and artery/vein classification under Prof. Tapan Kumar Gandhi, covering preprocessing, augmentation, patch sampling, and mask generation.
- Trained and benchmarked encoder-decoder segmentation architectures including Attention U-Net, tuning loss functions and hyperparameters against IoU and Dice coefficient.

DJUBO

Jan 2026 – Apr 2026

Software Developer Intern | Repository

Noida, India

- Built a large-scale employee attendance management platform using AngularJS, Django REST Framework, SQLite, and token-based authentication.
- Implemented REST endpoints powering dashboards, calendar views, admin workflows, leave management, and analytics visualizations via Handsontable and Highcharts.

Projects

Parkinson's Detection from Voice with Explainable AI | Python, Scikit-Learn, XGBoost, SHAP, LIME

GitHub

- Found that the usual way of splitting this dataset puts the same speaker in both training and test sets, letting models score well by recognizing a familiar voice rather than by detecting the disease.
- Re-tested 12 models over 960 runs with speaker-wise splitting, showing scores are inflated by 0.159 ROC-AUC on average and that the two methods disagree on which model is best.
- Used SHAP and LIME on unseen speakers to show one voice measure scores 0.924 ROC-AUC against 0.847 for all 16 combined, reducing the model to a two-measure screening test at 92% sensitivity.

Real-Time Attention Monitoring System | Python, OpenCV, MediaPipe

GitHub

- Built a tool that reads a live webcam feed and scores how attentive the person on screen appears, updating frame by frame.
- Extracted face landmarks, head pose, and upper-body posture per frame using OpenCV and MediaPipe, then combined these signals in a rule-based scoring engine that flags distraction.

Hisaab: Group Trip Expense Manager | React, TypeScript, Supabase, Tailwind CSS

GitHub | Live

- Built and deployed a web app that tracks who paid what on a group trip and works out the smallest set of payments that settles everyone, handling uneven splits and bills paid by more than one person.
- Stored money in whole paise and distributed rounding remainders one at a time, so every split adds back to the exact bill total.
- Made the app work offline by saving to the browser first and syncing to Postgres in the background, with row-level security and database-side checks that reject invalid writes.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript

AI / ML: TensorFlow, Scikit-Learn, XGBoost, OpenCV, MediaPipe; Deep Learning, Computer Vision, Image Segmentation

Libraries: NumPy, Pandas, Matplotlib, SHAP, LIME

Full Stack: React, AngularJS, Django REST Framework, REST APIs, Tailwind CSS, Supabase, SQL

Tools: Git, GitHub, Jupyter, Google Colab

Education

VIT Bhopal University

2023 – 2027

B.Tech in Computer Science Engineering (AI & ML), CGPA: 8.34 / 10.0

Bhopal, India

GD Goenka Public School, CBSE, Class XII: 78.2% | Class X: 87.8%

Certifications: Applied Machine Learning, University of Michigan | Cloud Computing, NPTEL | Internet of Things, NPTEL

Leadership & Achievements

Microsoft Technical Club, VIT Bhopal

Dec 2024 – May 2026

Vice President (prev. Team Lead, Event Management; Core Member, Technical Team)

Bhopal, India

- Led planning and delivery of SIH Success Blueprint, coordinating Smart India Hackathon finalists, speaker onboarding, and promotions.

VIT Bhopal x Johns Hopkins University HealthHack

February 2026

Team Leader, Finalist | Repository

Bhopal, India

- Led a team of 6 building CareFast, a mobile-first emergency healthcare application, advancing to the Final Round.